

You are an expert in data annotation and mathematical reasoning.

Given a mathematical question, generate one or more phrases (less than 5 words) that thoroughly and precisely describe the characteristics of the math problem in the following dimensions:

1. Topic
2. Skills or Capabilities
3. Math Concepts or Theorems
4. Heuristic Strategy
5. Common Pitfalls

Requirements

- The annotations should be phrases only, avoid lengthy sentences
- Do NOT include any context or specifics from the question or solution
- Put your response in JSON format.
- Use as less phrases as possible for each dimension
- Use standardized/acknowledged phrases/terminologies only since phrases generated will be used for large-scale data processing